

NewsBot Intelligence System 2.0 - Executive Summary

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What it does. An enhanced news analysis tool that goes beyond categorization, it also identifies themes (topics), compiles articles, assesses sentiment, processes non-English requests via translation, finds related articles with minimal effort, and includes a mini-interface to play with concepts on the fly.

How it works. The BBC-style cleaned, and label-completed dataset is compiled to train a fast classifier based on TF-IDF features. In addition, it includes:

- Topic Modeling (NMF and LDA) to identify most consistent topics
- POS & Sentiment (universal tagset + VADER) to note structure and emotions
- Extractive Summaries for lengthy pieces to be skimmed over
- Semantic Search for rapid, relevant responses
- Multilingual Pass for short foreign language text to be translated/analyzed
- Interactive Gradio UI for real time classification/sentiment/search

Why it matters. Real news streams are multilingual, chaotic, and urgent. Editors need more than one genre: they need patterns, precis, and access to similar background information in a snap. This project represents a compact version of that stream that's reproducible, timely, and easy to assess.

Key takeaways.

- Clean outputs you can verify quickly (metrics.json, confusion matrix, topics, summaries, search results, multilingual table)
- Lightweight methods that perform well without GPUs
- A single notebook that runs end-to-end and documents each step

What's next. Sentence-level embeddings for retrieval, a small-scale abstractive summarizer in the mix, and named-entity recognition and linking.

Final thoughts. An accessible, cohesive system that a teacher can execute, and a stakeholder can interpret - designed to pass a final and serve as portfolio support.